5.4L COMPRESSED NATURAL GAS (CNG) HEATER HOSE/FUEL RAIL ABRASION

AFFECTED VEHICLES: **1997 THROUGH 2000 F-SERIES VEHICLES EQUIPPED WITH A 5.4L CNG ENGINE**

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OVERVIEW

The affected vehicles will require the installation of updated heater supply and return hoses. When correctly installed, the ***new*** design of the heater hoses will prevent any contact between the braided flexible line of the fuel rail and the heater hoses. An inspection will be performed to determine if the braided flexible line has been damaged from prior contact with the heater hoses. The severity of damage will determine if a replacement fuel rail must be installed in addition to the heater hose kit. This procedure gives detailed instructions for the inspection and replacement of the fuel rail, and replacement of the heater hoses. For proper venting and leak testing of the fuel lines (if required), you will be referred to the Workshop Manual. Also, an authorized modifications label must be completed and affixed to the vehicle.

INSPECTION

1. Inspect the braided flexible line of the fuel rail for any abrasion caused by contact with the heater supply or return hoses. See Figure 1.
   - If there are no areas of abrasion on the braided flexible line, install the applicable replacement heater hose kit, following the instructions in this document.
   - If there are areas of abrasion found on the braided flexible line, inspect them closely to determine the severity of the damage and proceed as follows:

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- If the braided covering of the flexible line has minor abrasions that have not worn through to the shiny inner surface of the flexible line, the fuel rail does not require replacement. Install the applicable replacement heater hose kit, following the instructions in this document.
- If the braided covering of the flexible line shows severe abrasions that have worn through to the shiny inner surface of the flexible line, the fuel rail requires replacement. Install a replacement fuel rail with ***new*** fuel injector O-rings and the applicable replacement heater hose kit, following the instructions in this document.

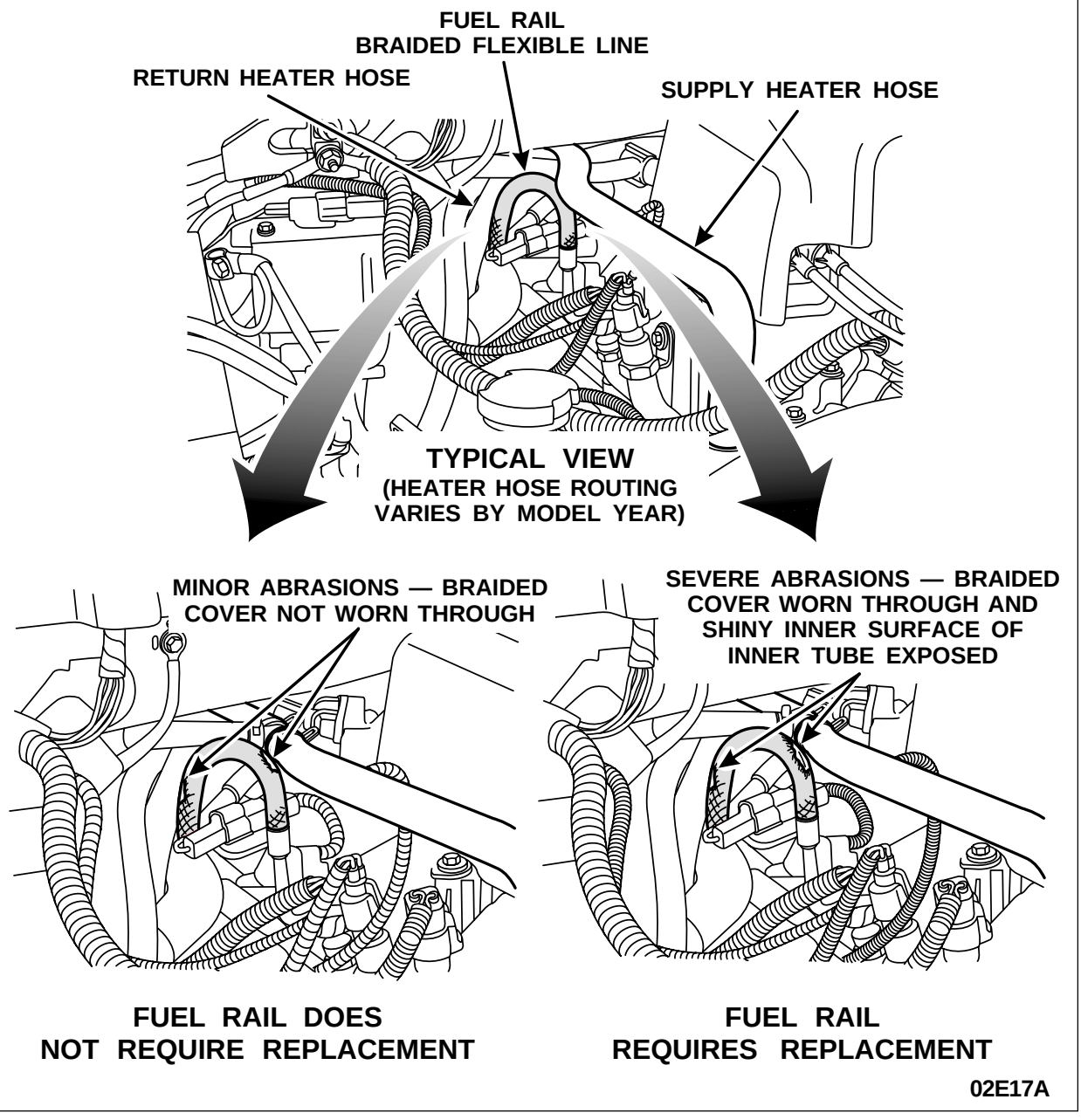


FIGURE 1

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HEATER HOSE AND FUEL RAIL REPLACEMENT

NOTE: The fuel rail requires replacement only if the inspection of the braided flexible line found any severe abrasions that have worn through to the shiny inner surface of the flexible line.

1. If the fuel rail requires replacement, you must relieve the pressure from the fuel lines at this time. Follow the Pressure Relief-Fuel Line Venting procedure in Section 310-00B of the appropriate Workshop Manual.
2. Install a memory saver, then disconnect the battery negative cable.
3. **WARNING: TO AVOID PERSONAL INJURY, DO NOT UNSCREW THE COOLANT PRESSURE RELIEF CAP WHILE ENGINE IS OPERATING OR HOT. THE COOLING SYSTEM IS UNDER PRESSURE; STEAM AND HOT LIQUID CAN COME OUT FORCEFULLY WHEN THE CAP IS LOOSENED SLIGHTLY.**

   CAUTION: The coolant must be recovered in a suitable, clean container for reuse.

   Drain the engine cooling system.
4. Disconnect the positive crankcase ventilation (PCV) valve from the right side valve cover, then position the valve and line out of the way.
5. For 1997 through 1999 vehicles, reposition the electrical junction block located on the dash panel as follows. See Figure 2.
   - Remove the top mega fuse cover.
   - Disconnect the positive cable from the mega fuse stud.
   - Remove the screws from the electrical junction block bracket and position the junction block out of the way.

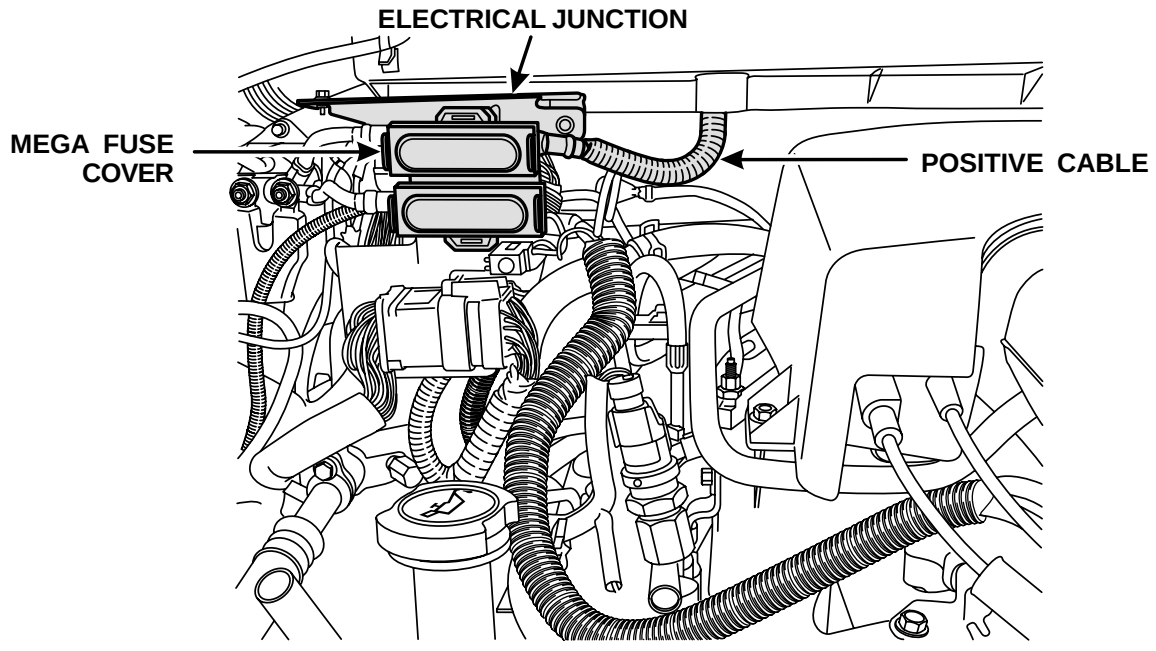


FIGURE 2

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6. Disconnect and remove the coil-on-plug from the rearmost cylinder on the right side of the engine.
7. NOTE: The plastic retainer clip may remain on the heater core nipple. If this occurs, the clip must be removed and discarded. The ***new*** hose assembly comes with a ***new*** clip.

   Remove the heater supply and return hoses. To release the hose connections at the heater core, press the retainer clip tabs inward and pull the hose from the heater core nipple. See Figure 3.

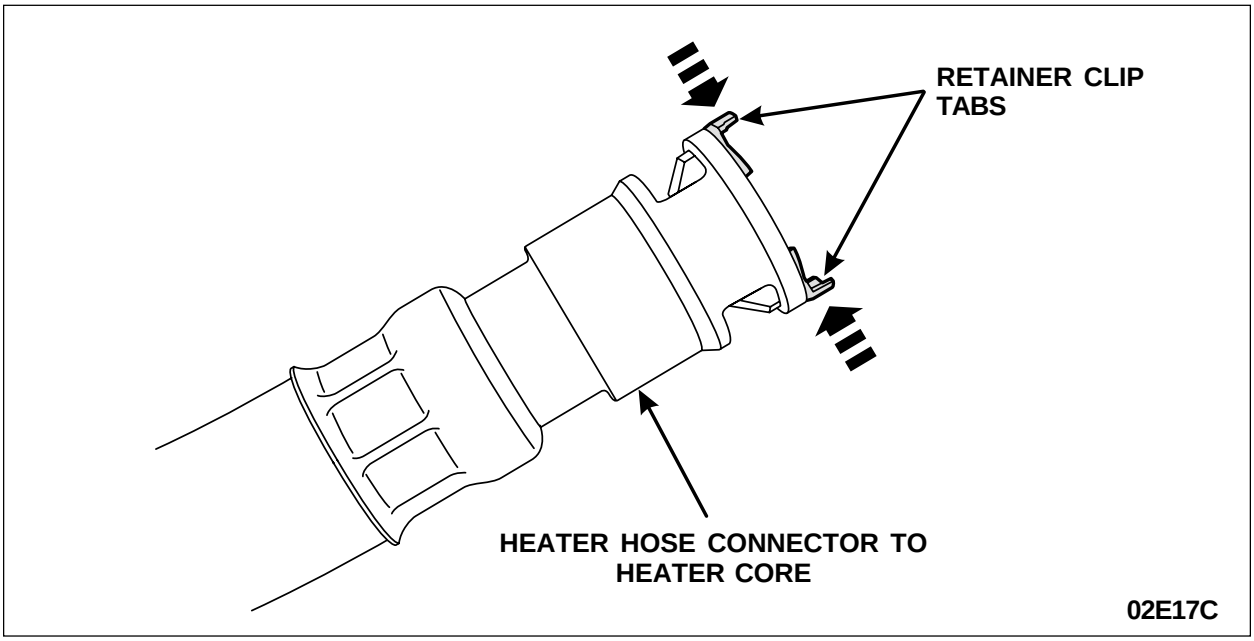
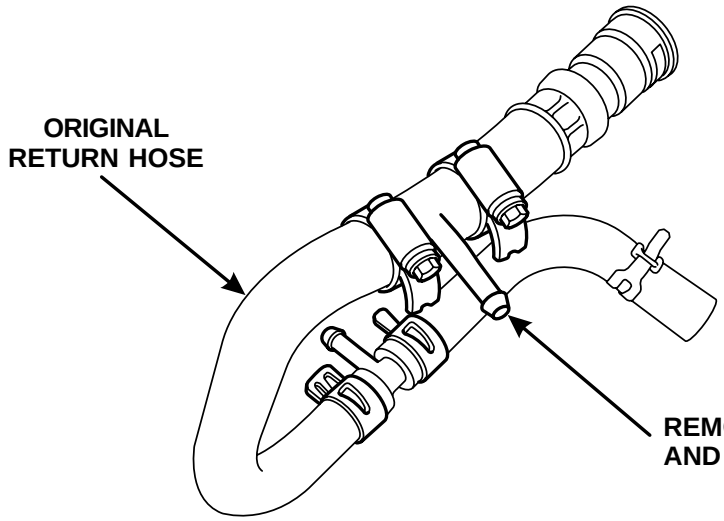


FIGURE 3

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8. NOTE: The heater return hose on some vehicles may have been modified with a second tee fitting. This is for a heated throttle body adapter, and will be installed onto the ***new*** return hose.

   If the vehicle has been modified with a heated throttle body adapter, remove the water tee fitting and clamps from the heater return hose. See Figure 4.



VEHICLES WITH HEATED THROTTLE BODY ADAPTER ONLY
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FIGURE 4

NOTE: Perform steps 9 through 27 only on vehicles that require a *new* fuel rail.

9. Remove the throttle body cover.
10. Disconnect the intake air temperature (IAT) and mass air flow (MAF) sensors.
11. Disconnect the air cleaner outlet tube from the throttle body, then remove the air cleaner outlet assembly.

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12. Disconnect the following electrical connectors. See Figure 5.
    - Idle air control (IAC) valve
    - Injection pressure sensor
    - Engine fuel temperature sensor
    - Fuel shutoff valve
    - All eight (8) fuel injectors
    - Left and right radio capacitors (if equipped)
13. Disconnect the vacuum line from the side of the throttle body adapter. See Figure 5.

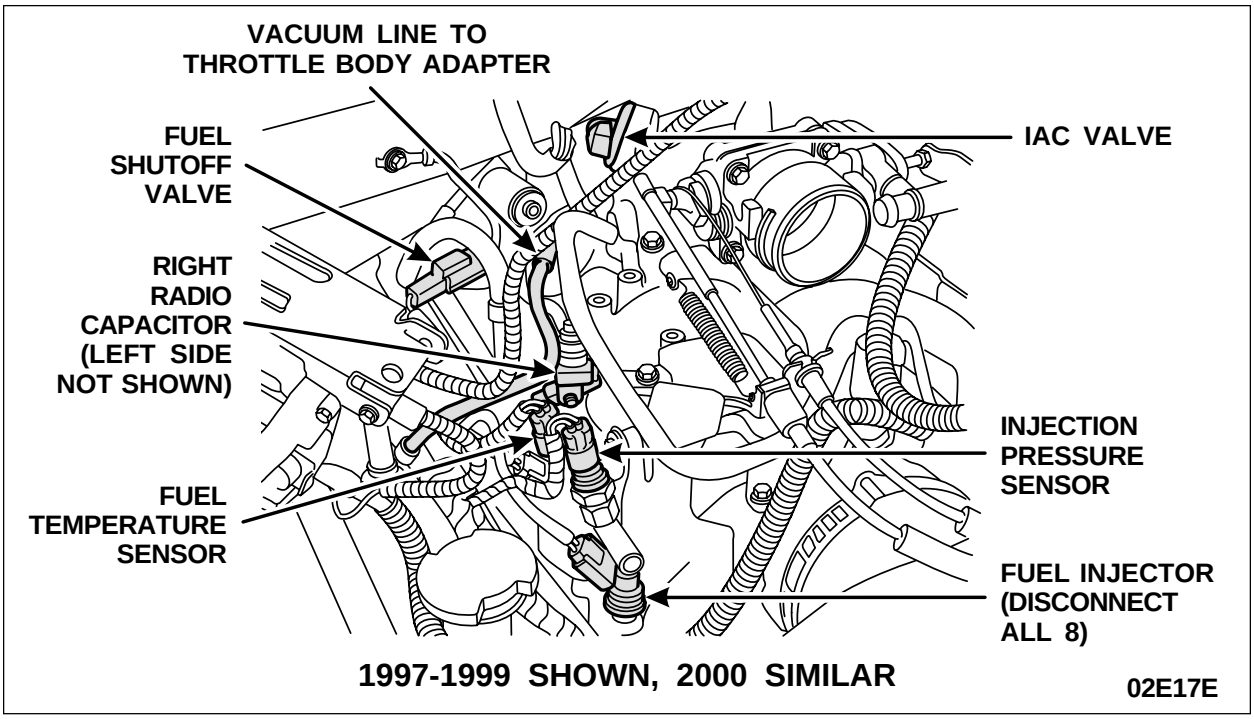


FIGURE 5

14. Remove the safety clip from the fuel line connection at the fuel rail.

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15. Disconnect the fuel line from the fuel rail, using a suitable spring lock coupling tool. See Figure 6.
16. Remove the bolts from the fuel rail, then lift the fuel rail with the fuel injectors attached from the engine. See Figure 6.

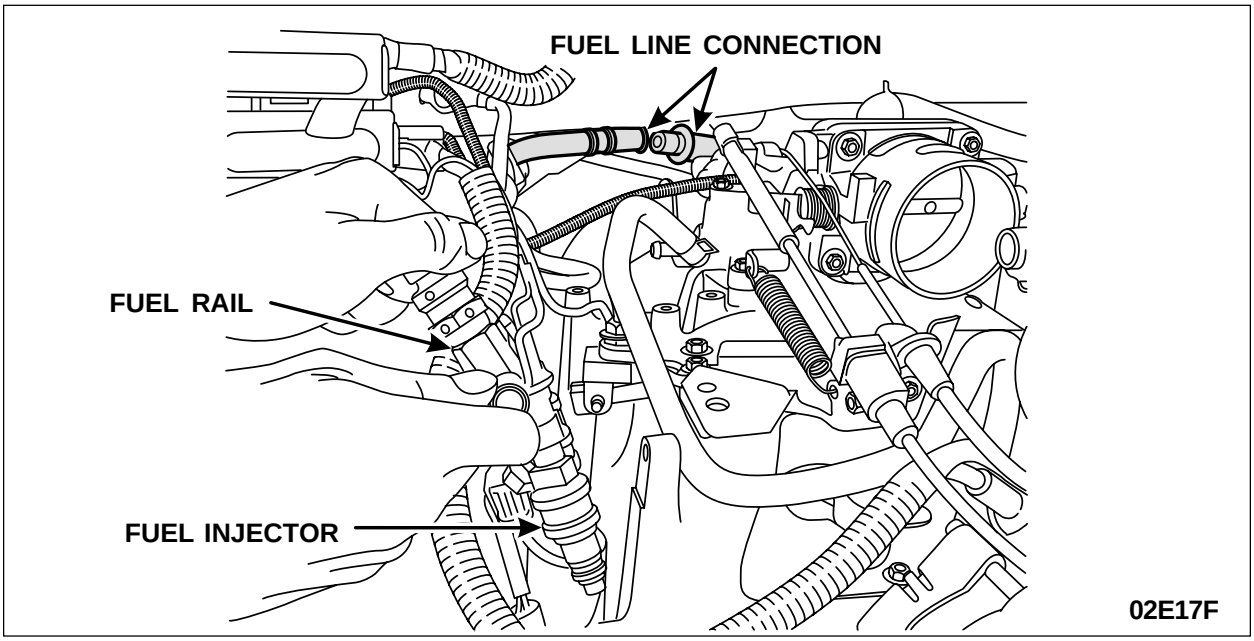


FIGURE 6

17. Separate the fuel rail from the fuel injectors, then destroy and discard the fuel rail.

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18. **CAUTION: The fuel injector O-rings for the CNG engines are made of a different material and color (yellow) than those for the gasoline engine. Therefore, the yellow fuel injector O-rings included in kit F3DZ-9229-AAFFV are the only O-rings that should be used.**

    Remove and discard the fuel injector O-rings. Then, install two (2) ***new*** O-rings on each fuel injector, using O-ring kit F3DZ-9229-AAFFV and lubricate them with 5W30 engine oil meeting Ford specification WSS-M2C153-G. See Figure 7.

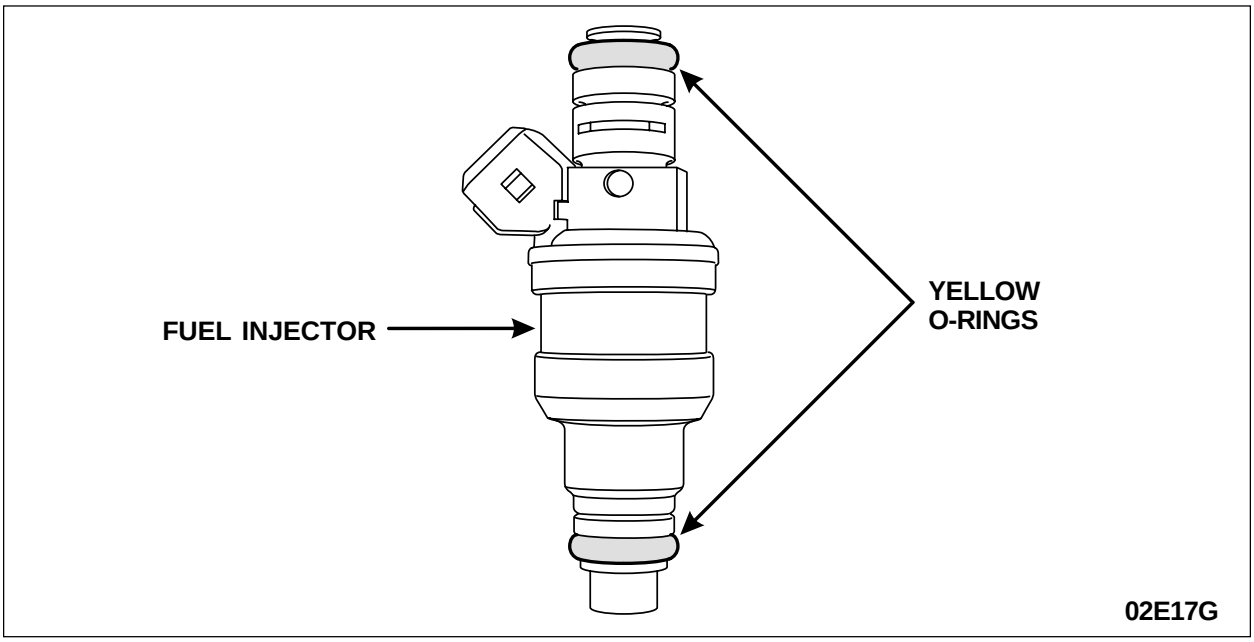


FIGURE 7

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19. NOTE: Make sure the fuel injector mounting holes are free from debris prior to installing the injectors.

    Install the fuel injectors into the engine.

20. NOTE: To ensure proper fit, do not connect the fuel rail to the injectors during this step.

    Position the ***new*** fuel rail over the engine, then connect the fuel line and install the safety clip. See Figure 8.

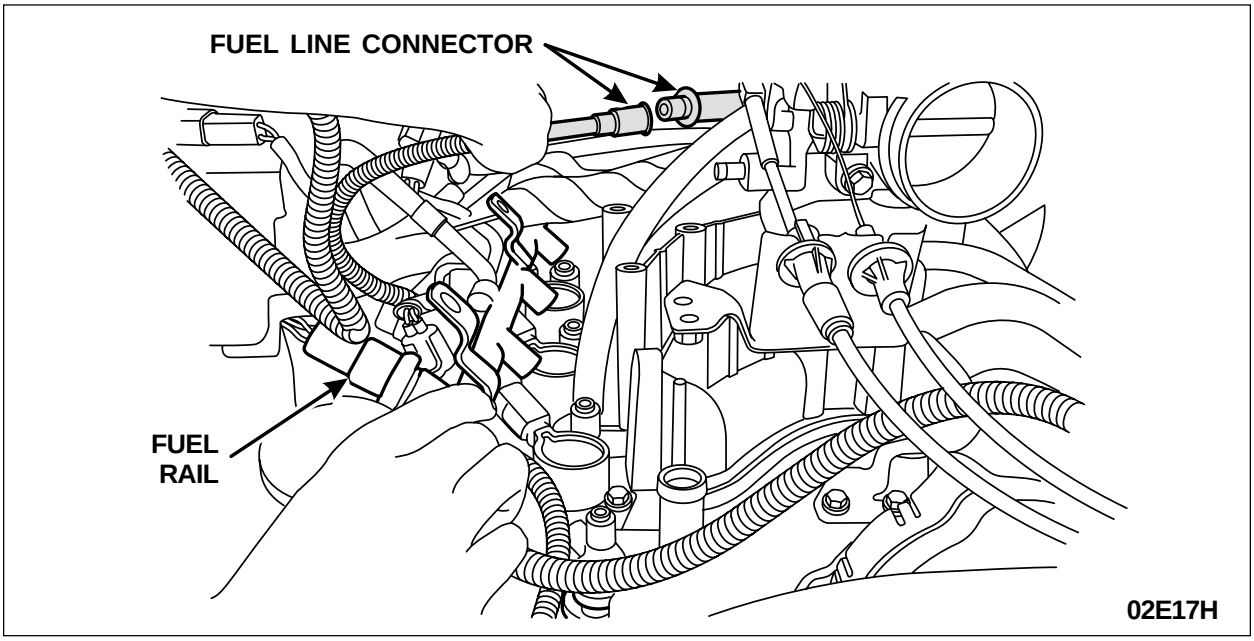


FIGURE 8

21. **CAUTION: The yellow fuel injector O-rings are soft and easily damaged. Do not roll or rock the fuel rail onto the fuel injectors.**

    Position the fuel rail squarely onto the fuel injectors. Press the fuel rail straight down onto all eight (8) fuel injectors.

22. Install the fuel rail bolts. Tighten the bolts to 10 Nm (89 lb-in).
23. Connect the vacuum line to the side of the throttle body.
24. Connect the following electrical connectors.
    - Left and right radio capacitors (if equipped)
    - All eight (8) fuel injectors
    - Fuel shutoff valve
    - Engine fuel temperature sensor
    - Injection pressure sensor
    - Idle air control (IAC) valve
25. Connect the air cleaner outlet tube to the throttle body, then install the air cleaner outlet assembly.
26. Connect the intake air temperature (IAT) and mass air flow (MAF) sensors.

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27. Install the throttle body cover.
28. NOTE: Do not final tighten the tee fitting clamps at this time.

    If the original hose was equipped with a heated throttle body water tee fitting, install the tee fitting on the ***new*** hose as follows. See Figure 9.
    - Mark the ***new*** heater return hose as shown in the illustration. This mark will be used for aligning the hose prior to tightening the water tee clamps.
    - Cut the heater hose as shown and loosely install the water tee fitting and clamps.

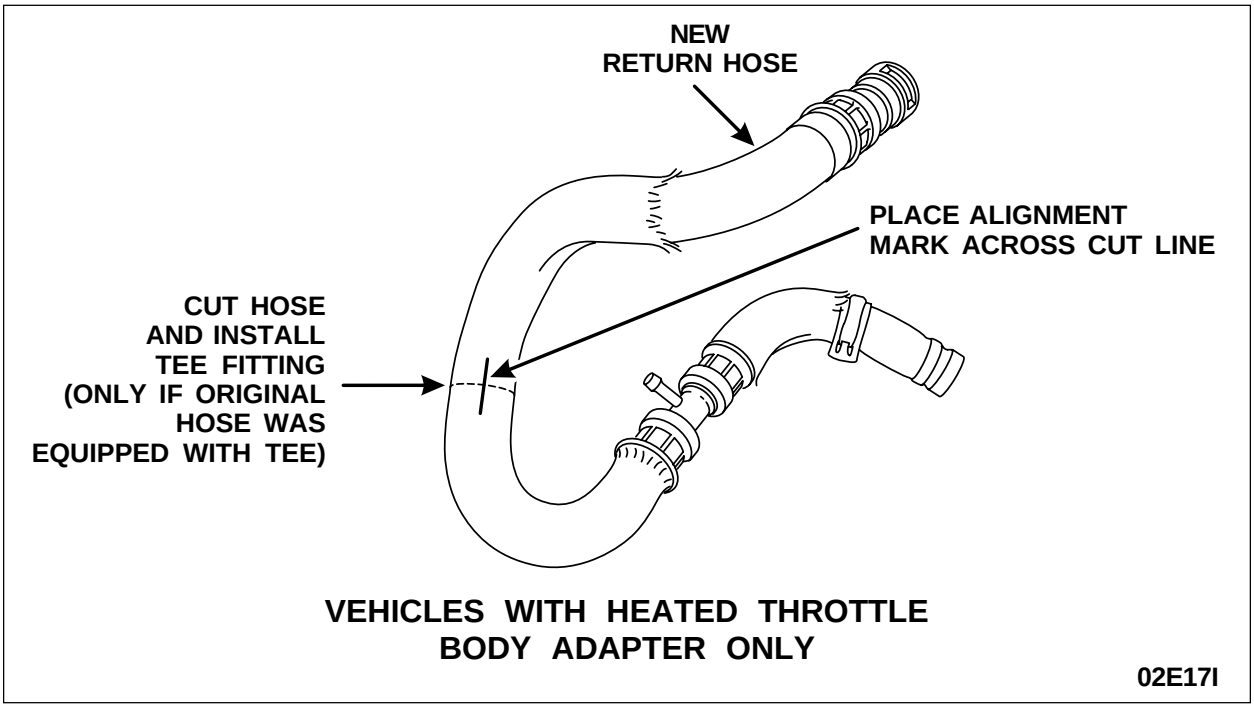


FIGURE 9

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29. NOTE: For proper routing, the length of the fuel regulator coolant hoses on 1997-1999 vehicles must be modified. This is accomplished by transferring the extension hose on the fuel regulator coolant hoses.

    On 1997-1999 vehicles, transfer the extension hose and clamps from the fuel regulator coolant return hose to the fuel regulator coolant supply hose. See Figure 10.

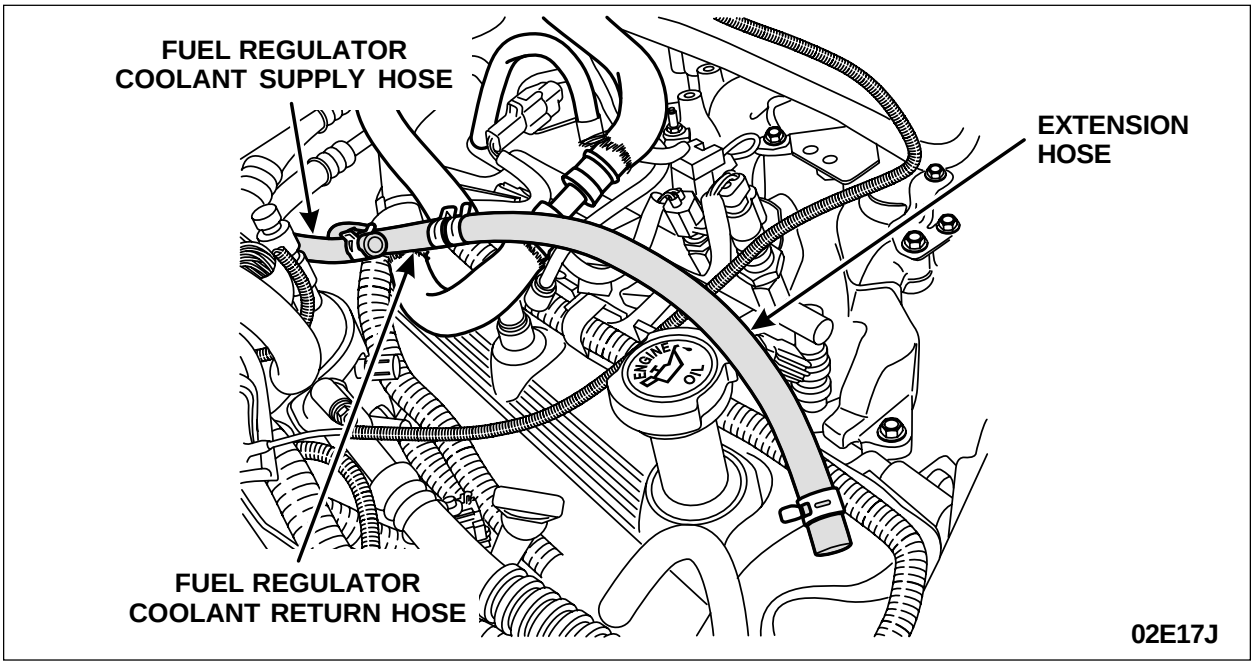


FIGURE 10

30. **WARNING: DO NOT ROUTE ANY HOSES UNDER OR NEAR THE BRAIDED FLEXIBLE LINE OF THE FUEL RAIL.**

    Install the ***new*** heater supply and return hoses. Make sure the hoses are routed as shown in the illustration. Do not route any of the fuel regulator coolant (or heated throttle body, if equipped) hoses under or near the flexible line of the fuel rail. See Figure 11 for 1997-1999 vehicles or Figure 12 for 2000 vehicles.

31. If a heated throttle body water tee fitting is installed on the return hose, proceed as follows.
    - Align the reference marks on the return hose.
    - Tighten the water tee fitting clamps.

32. Install the supply heater hose routing clip as follows. See Figure 11 for 1997-1999 vehicles or Figure 12 for 2000 vehicles.

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- **For 1997-1999:**
  - Install the P-clip included in the kit around the ***new*** heater supply hose and onto the intake manifold stud.
  - Install the nut included in the kit onto the stud and tighten to 10 Nm (89 lb-in).

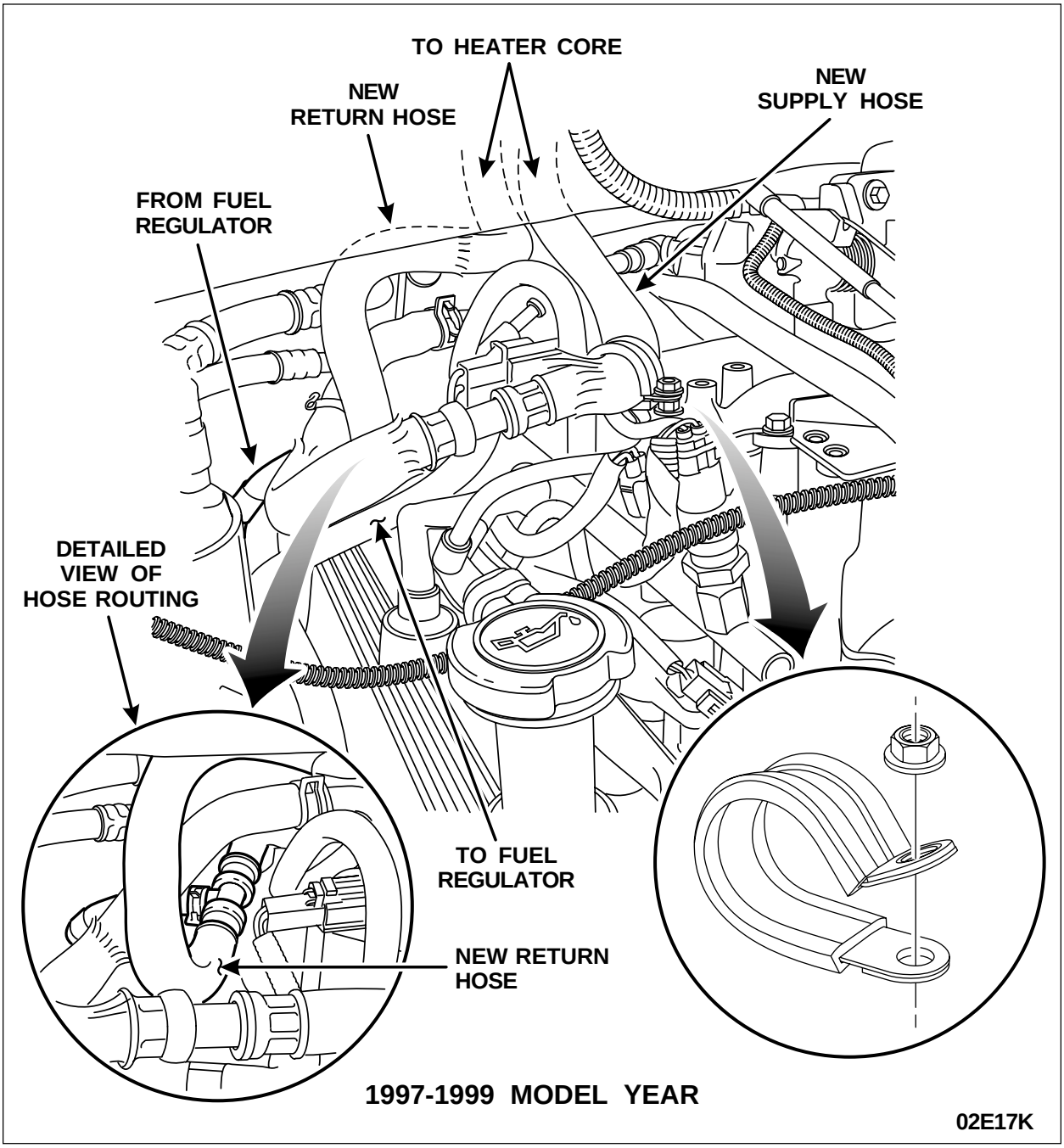


FIGURE 11

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- **For 2000:**
  - Install the stud included in the kit into the mounting boss on the intake manifold. Tighten the stud to 10 Nm (89 lb-in).
  - Slide the retaining bracket included in the kit onto the stud. The opening of the bracket must point toward the engine cover as shown in the illustration.
  - Snap the retaining bracket closed to secure the supply hose.

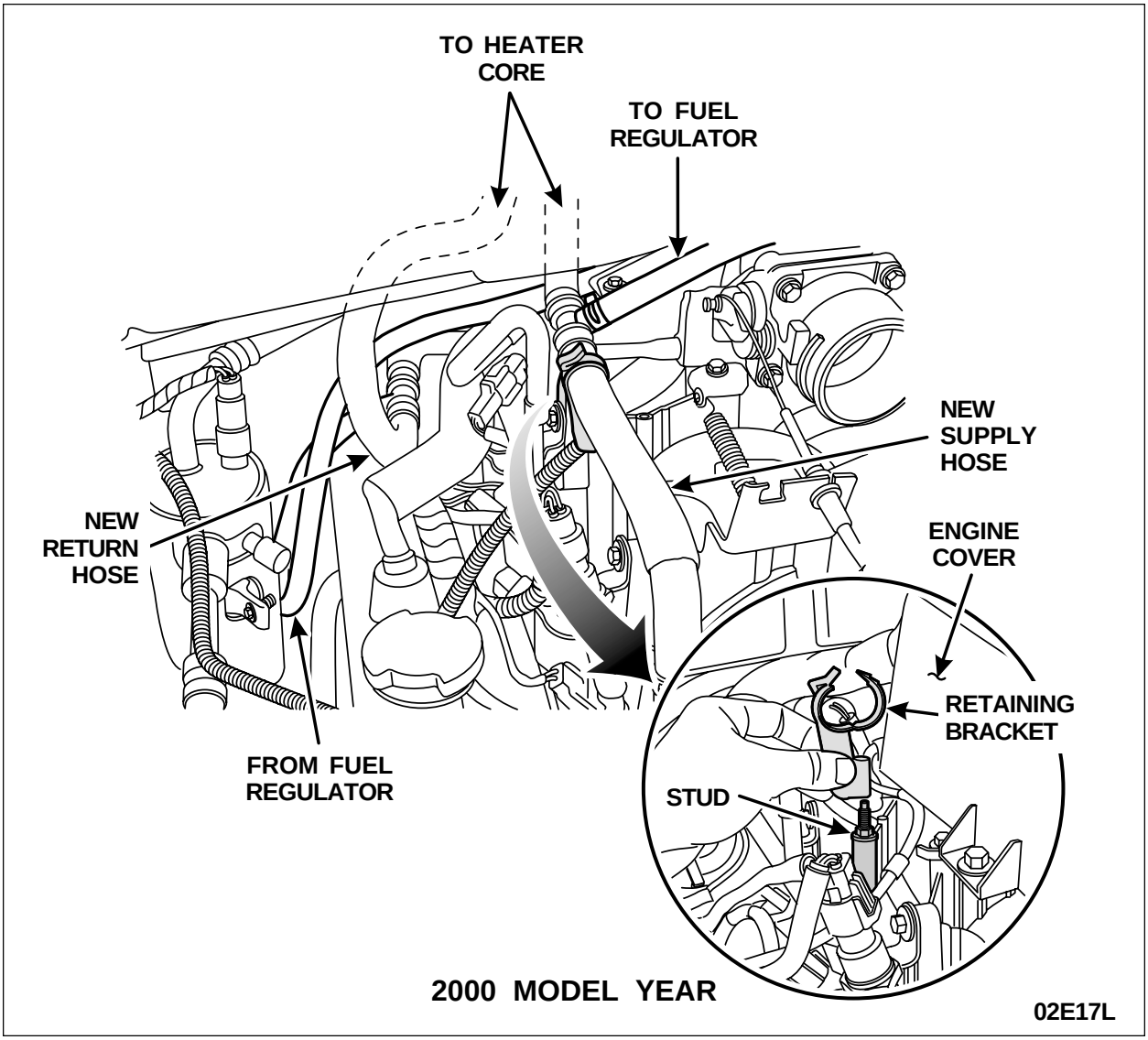


FIGURE 12

33. Install and connect the coil-on-plug for the rearmost cylinder on the right side of the engine.

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34. For 1997 through 1999 vehicles, install the electrical junction block located on the dash panel as follows.
    - Position the junction block bracket onto the bulkhead and install the screws.
    - Connect the positive cable to the mega fuse stud.
    - Install the mega fuse cover.
35. Connect the PCV valve into the right side valve cover.
36. Connect the battery negative cable and remove the memory saver.
37. If the fuel rail was replaced:
    - Turn out the manual lockdown jackscrews on the fuel tanks (jackscrews were turned in during fuel line venting).
    - Perform a fuel system leak test. Refer to the procedure in Section 310-00B in the appropriate Workshop Manual.
38. Fill the cooling system through the degas bottle.
39. Set the heater blend control to the full warm position. Start the engine and allow it to idle until normal operating temperature is reached. Hot air should discharge from the A/C vents. The engine coolant temperature gauge should maintain a stabilized reading in the middle of the NORMAL range and the upper radiator hose should feel hot to the touch.
40. Shut the engine off and allow it to cool.
41. Check and top off the degas bottle with coolant.

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PREPARE AUTHORIZED MODIFICATIONS LABEL

1. Obtain and prepare the Authorized Modifications label. Write the modification completed, your P&A code, date of repair and the recall program number in the spaces provided on the label. See Figure 13.

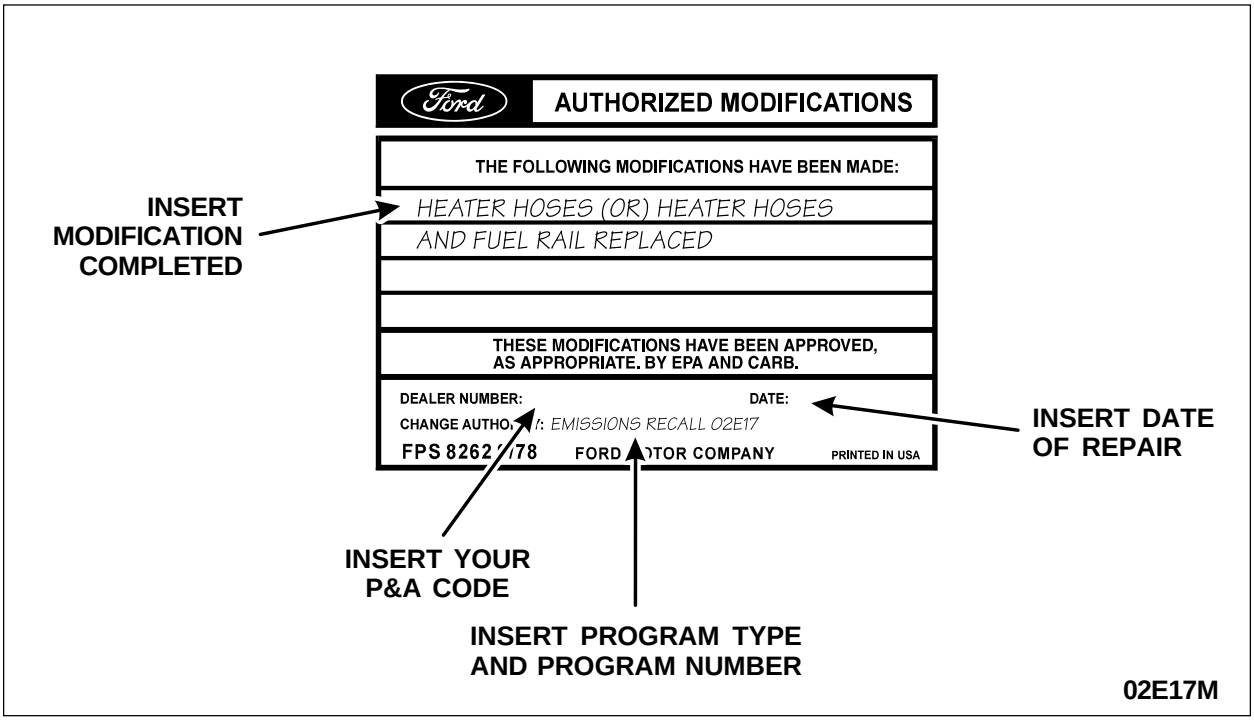


FIGURE 13

2. Clean an area of the radiator support with isopropyl alcohol and a clean cloth.
3. Apply the label to the clean area.

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4. Provide California owners with a completed Vehicle Emissions Recall–Proof of Correction certificate. See Figure 14.

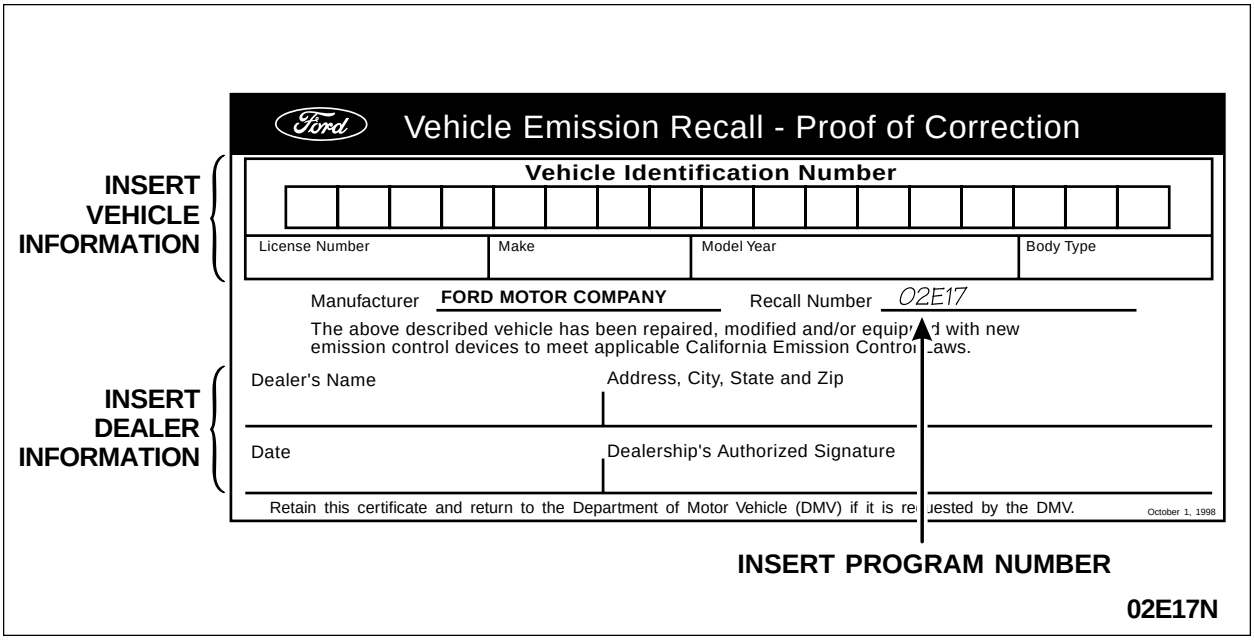


FIGURE 14

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